CHIP-BOY Mk III Reconstruction and verification report

November 3, 2026 | Rev A | Original through-hole components retained

Overall result: geometry recovered; electrical release incomplete

This report documents the source audit, mechanical reference and through-hole reconstruction performed to date. It does not certify a completed EasyEDA PCB. No substituted modules have been adopted. Native EasyEDA import, complete electrical pin mapping, routing and manufacturing verification remain unfinished.

Work item	Result
Source preservation	50 supplied files checked against their saved SHA-256 fingerprints; no source edits.
Mechanical reference	92.710 x 63.500 mm, five closed contours. Shared outline and mounting features agree with the bottom mechanical drawing within 0.016510 mm after recorded repairs.
Through-hole geometry	24 groups exported: 138 round component/locator holes plus two plated slots. 51 small holes remain classified as via candidates.
Source layer comparison	Recovered component pad shapes/dimensions checked against top and bottom copper; corresponding mask openings compared on both sides.
Independent CAD read-back	24 groups, 138 round pads and two slots successfully parsed. No duplicate round-hole matches; no errors at 0.000002 mm comparison tolerance.
Electrical mapping	85 pads across 11 groups have reconstructed mappings. These are not independently certified against an original native netlist.
EasyEDA Pro	Existing draft project remains. Connection timed out during this session; import/save could not be verified.

Files are engineering review artifacts, not a BOM approval, complete device library or fabrication release. Physical part identity, body clearance and fit are not proved by matching drill geometry.

1. Source basis and mechanical checks

Evidence used

Read-only source: C:/Users/reahs/Downloads/CHIP-BOY-MK.-3-main/CHIP-BOY-MK.-3-main. The reconstruction uses the supplied schematic PDF, assembly drawing, mechanical DXFs, GTL/GBL copper, GTS/GBS masks, GM1 outline, Excellon drill files and DRR drill report. Native Altium source documents and firmware are absent.

The Gerber ZIP contains 37 members. Its integrity check passed; member contents match extracted outputs after CRLF/LF normalization. Byte differences caused only by line endings were not treated as design changes. All six mechanical DXFs parsed without audit errors or automatic fixes.

Original design constraints

The extension report identifies four copper layers: top, internal GND, internal 3V3 and bottom. The generated review board uses two copper layers solely to carry recovered pad geometry; it is not a recreation of the original stackup or routing. Do not manufacture the review board.

Original DRC reports zero violations but includes overlapping all-object 10 mil and 0 mil clearance rules and zero silkscreen clearances. Its result does not establish suitable redesign rules or certify these reconstructed footprints. The DRC/output date precedes the schematic date, so source revision correspondence is not established.

Mechanical reference details

Feature	Recovered reference
Envelope	92.710 x 63.500 mm
Mounting holes	Nominal diameter 3.2512 mm. Centers: (14.478, 4.826), (89.916, 11.557), (51.435, 60.706) mm.
Coordinate system	Origin at lower-left envelope; X right, Y up. Original CAM offset: (57.658, 50.800) mm. Review-board coordinates deliberately include a drawing offset and invert Y.
Internal slot	GM1 includes an internal slot absent from the bottom enclosure DXF. It was retained as a PCB reference feature.
Geometry repairs	Maximum source endpoint gap 0.033020 mm; merged endpoints can move 0.016510 mm. Curves are polygon approximations with a target chord error of 0.003 mm.

Correction to the earlier audit: TOF400C and VL53L1X are not necessarily conflicting descriptions. The module name can refer to a breakout using that sensor. Exact board identity is still not established from the names alone.

2. Reconstructed component mappings

These 11 groups contain 85 round pads. Pad numbers and functions were reconstructed from schematic symbols, assembly labels and physical ordering. The CSV and JSON preserve original hole IDs, source coordinates, local coordinates, drills, pad shapes and mask margins. Geometry checks passed; complete original-netlist equivalence has not been demonstrated.

Reference	Component/interface	Pads	Mapping basis / constraint
M1	XIAO carrier interface	14	D0-D10, 3V3, GND, 5V; original two header rows retained.
U1	MCP23008	18	DIP arrangement and schematic functions; no package substitution.
U2	DFPlayer	16	Original two-row interface retained.
U3	GPS	5	Original leading-zero pin identifiers 01-05 retained.
U4	BNO055	10	Original header arrangement retained; no STEMMA QT substitution.
DS	OLED	4	GND, VCC, SCL, SDA mapping retained.
J5	BME280	6	Upper six-pin header; source pin order retained.
J4	TOF400C	6	Right-hand six-pin header; VCC is pin 6. Square pad does not imply pin 1.
J1	Battery	2	Original battery connection and polarity labels.
J2	ESP battery leads	2	Original GND/VCC numbering; not interchangeable with J1.
J3	Laser connection	2	Original supply/return mapping.

Verification performed

Each exported round-pad center was checked against the drill data. Top and bottom copper apertures and both solder-mask apertures were compared before export. An independent KiCad parser reopened the board, then measured pad positions, round-pad dimensions, drill diameters and slot geometry against the recovered source. All 138 round pads matched distinct source holes. Maximum numerical discrepancy was below 0.000002 mm; this measures file conversion fidelity, not manufacturing tolerance.

Library limitations

The footprints currently contain pads and identification text. They do not reconstruct complete original body outlines, courtyards, 3D models or library metadata. Symbols/devices are not bound and validated inside EasyEDA. No electrical netlist or traces are included in the review board.

3. Remaining groups and explicit holds

All remaining hole geometry has been exported into a separate reference-only library. Pads with unknown electrical identity retain source hole IDs instead of invented terminal numbers. These held groups must not be used as released schematic-connected footprints.

Group	Recovered	Required evidence before release
RE1 encoder	Five round electrical holes and two plated mounting slots.	Verify A/B/common and push-switch continuity; confirm shield pad numbering.
S1-S5 switches	Five six-lead patterns, 30 holes.	The schematic shows three terminals per switch. Establish physical contact grouping and mapping of the second pole from the original component drawing, native design or continuity measurements.
LED1-LED3	Three two-lead patterns, six holes.	Confirm electrical terminal numbering and cathode/anode mapping against original part or traced net connectivity.
Two-pin audio region	Two-hole footprint retained at source location.	Assembly annotations do not conclusively identify whether this region represents LS1 or BZ1. Resolve identity and polarity before assigning a reference.
B0 and B1	Two four-lead tactile patterns, eight holes.	Verify internally common contact pairs and source terminal numbering.
B2 navigation locators	Two non-plated locating holes.	The navigation switch is surface-mount. Its six electrical pads are outside this through-hole-only reconstruction and remain to be rebuilt.

Related electrical issues remain active

The original D8/D9/D10 allocation conflicts with the XIAO Sense microSD bus according to the earlier source review. The BNO055 vendor also documents ESP32-S3 I2C interoperability concerns. These require architecture and firmware work without substituting the requested modules. Battery charge/discharge limits, laser current, speaker load and buzzer drive are not established by a footprint.

Evidence needed to close the holds

A matching original .PcbDoc/.SchDoc or exported pin/net list would establish source terminal identities. Where unavailable, original component drawings and continuity/polarity measurements can establish the electrical mapping. Additional copper-connectivity reconstruction may resolve some holds, but has not been completed here. No physical measurements or bench tests were performed.

4. Deliverables, EasyEDA status and next actions

File	Purpose
through-hole-review-package.zip	24 geometry groups, review board and machine-readable checks; contains both mapped and held libraries.
complete-geometry-review.kicad_pcb	Combined positional review board in KiCad 5.1 syntax for conversion testing. Not a routed circuit.
OriginalCarrier.pretty	11 reconstructed component-interface footprints.
UnresolvedOriginal.pretty	13 held reference groups with source hole IDs where terminal identity is unknown.
reconstructed-pin-map.csv / .json	85-pad mapping and source geometry evidence.
remaining-footprint-checks.json	Held groups, checks and unresolved reasons.
independent-footprint-verification.json	Independent parser counts and measurement comparison.
source-manifest.json	Original source fingerprints.

EasyEDA status

Draft project: https://pro.easyeda.com/editor#id=987575100d2f4542bfc9780160616df2. In this session, the tab was visible in the browser inventory, but attaching to it timed out. No import or save was performed or verified. Previous successful work in that project was scaffolding and a draft note, not a completed schematic or PCB.

Required work before declaring Step 3 complete

1. Resolve the held electrical identities and original component numbers. 2. Finish missing surface-mount navigation geometry and component bodies/keepouts where required. 3. Import the library through EasyEDA supported conversion, bind each symbol to the intended footprint, and compare pad numbers and physical dimensions after conversion. 4. Save, reopen and verify the native project. 5. Perform 1:1 physical fit/polarity checks on actual modules.

Required work after Step 3

Finalize power architecture and GPIO allocation; complete schematic capture; place and route the carrier with appropriate stackup/rules; run ERC/DRC and manual checks; independently inspect manufacturing exports; prototype and bench-test. None of these later steps has been certified by this report.

Final assessment

Mechanical hole recovery and file read-back verification are complete for the scoped through-hole features. Full electrical mapping and EasyEDA integration are incomplete. The evidence supports using the package for engineering review only. A zero-error or manufacturing-ready claim would not be justified.